

Hamming Code.

(Correct error of a single bit).

$r = 8, 7, 6, 5$

Data $\Rightarrow m$

Redundancy bit $\Rightarrow r$

Total sent data $\Rightarrow m+r$

Data $\Rightarrow (m) \rightarrow 101100110011 \rightarrow 12 \text{ bit.}$
 $r \Rightarrow 5$

Formula $\Rightarrow 2^r \geq m+r+1$

$$2^8 \geq 12+8+1$$

$$\Rightarrow 2^4 \geq 12+4+1$$

$$16 \geq 17 \times$$

$$\Rightarrow 2^5 \geq 12+5+1$$

$$32 \geq 18 \checkmark$$

$$\begin{aligned} 2^0 &= 1 \\ 2^1 &= 2 \\ 2^2 &= 4 \\ 2^3 &= 8 \\ 2^4 &= 16 \end{aligned}$$

10001	10000	01111	01110	01101	01100	01011	01010	01001	01000	00111	00110	00101	00100	00011	00010	00001
17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1
r_{16}									r_8				r_4		r_2	r_1

Sent Data $\Rightarrow$

1101100110001111

$$r_1 = (3, 5, 7, 9, 11, 13, 15, 17) \\ = 1 \oplus 1 \oplus 0 \oplus 1 \oplus 0 \oplus 1 \oplus 0 \oplus 1 \Rightarrow 1$$

$$r_4 \Rightarrow 1 \\ r_8 \Rightarrow 0 \\ r_{16} \Rightarrow 1$$

$$r_2 \Rightarrow (3, 6, 7, 10, 11, 14, 15) \\ \Rightarrow 1 \oplus 0 \oplus 0 \oplus 0 \oplus 1 \oplus 0 \oplus 1 \oplus 0 \Rightarrow 1$$

Received Data: $\rightarrow$ 1 1 0 1 1 0 0 1 0 0 0 0 1 1 1 1

bit changed.

$\rightarrow$ 1 1 0 1 1 0 0 1 1 0 0 0 1 1 1 1

Corrected Code.

$$r_1 \Rightarrow 1 \\ r_2 \Rightarrow 0 \\ r_4 \Rightarrow 0 \\ r_8 \Rightarrow 1 \\ r_{16} \Rightarrow 0$$

$$\begin{array}{ccccccc} r_{16} & r_8 & r_4 & r_2 & r_1 \\ 0 & 1 & 0 & 0 & 1 \end{array} \Rightarrow 9\text{th bit is changed.}$$

Received Data! $\rightarrow$ 1 1 0 1 1 0 0 1 0 0 0 0 1 1 1 1 1
 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1

$r_1 \rightarrow 1, 3, 5, 7, 9, 11, 13, 15, 17,$

$$= 1 \oplus 1 \oplus 1 \oplus 0 \oplus 0 \oplus 0 \oplus 0 \oplus 1 \oplus 0 \oplus 0 \oplus 1 = 1$$

$r_2 \rightarrow 2, 3, 6, 7, 10, 11, 14, 15,$

$$\rightarrow 1 \oplus 1 \oplus 0 \oplus 0 \oplus 0 \oplus 1 \oplus 0 \oplus 0 \oplus 1 \oplus 0 \rightarrow 0$$

$r_4 \rightarrow 4, 5, 6, 7, 12, 13, 14, 15 \rightarrow 1 \oplus 1 \oplus 0 \oplus 0 \oplus 0 \oplus 0 \oplus 1 \oplus 1 \oplus 0 \rightarrow 0$

$r_8 \rightarrow 8, 9, 10, 11, 12, 13, 14, 15 \rightarrow 0 \oplus 0 \oplus 0 \oplus 1 \oplus 0 \oplus 0 \oplus 0 \oplus 1 \oplus 0 \rightarrow 1$

$r_{16} \rightarrow 16, 17 \rightarrow 1 \oplus 1 \rightarrow 0$

HW

practice all
the maths

VRC, LRC, CRC,

Check sum &

Hamming Code.

